

MLDawn "Brain in the Dark Design Principles for Neuromimetic Inference"

1:03:55

hello welcome back it's the fourth
applied active inference Symposium day
two part two on November 14th
2024 we're kicking off this session in
the darkness of the preon on the west
coast appropriately with Maron Hussein
Zade

bagani ml Dawn brain in-the-dark design
principles for the neuromatic inference
under the free energy principle so thank
you Maron for joining looking forward to
your

presentation thank you very much it's a
pleasure to be here

um so today I'm going to be talking
about uh the design principles for
neuromatic inference under the free
energy principle namely the methods that
are inspired by the brain um in order to
model human

perception um just um want to preface
this by saying that this is a

collaboration with my colleague Simon
orbas from Manu University and Professor
friston from UCL and this work has
recently been accepted at the neuro AI
workshop at neurs

2024 so let's just get

started the content is going to be as
follows we're going to be talking about
first the motivation behind this work
why we got started started with this
idea anyway to to introduce the design
principles particularly for these models
and then I'm going to do a quick

refresher on the math side just so we all get comfortable with the you know the upcoming derivations that I'm going to be uh expanding on which covers the math behind free the free energy principle particularly a variational free energy then I'll do a brief discussion on what free energy principle is I'm presume knows by now because we we' had so many talks uh since uh the beginning of the Symposium but I'm going to do a quick run at that then we'll be talking about the beian brain hypothesis that furnishes the foundation for the idea of modeling human perception as a kind of inference namely basan inference next we dive into dejos between the generative model and a generative process what do we mean by each one of them knowing the distinction is crucial for understanding the free energy principle because Fe lies on this um notion that there is a boundary that separates an entity from uh the environment and separating uh basically my internal state from the external State beyond my marov blanket next we'll talk about variational free energy and how it's driven how it's derived basically and then we jump into the design principles in terms of the experiments not the things that you need to consider the things that you need to Define and how you should go about doing that in order to model human perception as a kind of online inference problem we'll go through some final thoughts some lessons learned and then

I'll introduce some of the resources for those of you who are interested in uh you know pursuing free energy principle more seriously and delving into the theory and the Practical side of it and then we're going to wrap it up with uh a demonstration um using the GitHub repository that that is publicly available by um by the way so the motivation is this so you said deep learning has revolutionized artificial intelligence um because it has automated the process of feature extraction from raw data however it actually faces a lot of problems for for example lack of outof distribution generalizability catastrophic forgetting and poor interpretability it is known as a black box basically and the catastrophic forgetting is basically when the new knowledge overwrites the previous knowledge the weights are getting overwritten so while the network learns the concept of a cat it might momentarily forget the concept of a dog because the paths are shared between different Concepts that are to be learned you see in contrast to deep learning biological neural networks in your brain and a my brain they do not suffer from these issues and this inspires the AI researchers to actually explore this interesting and intriguing area called The neuromatic Deep learning that is all about replicating the brain mechanisms namely the neuronal message passing and belief updating within the AI models purely sprayed by our best

knowledge about how the brain works um a
foundational theory for this approach is
the free energy principle um Fe which
unfortunately despite its potential is
often considered too complex to
understand and to implement an AI simply
because one would need a myriad of
disciplines um under their belt
knowledge about disciplines under their
belt to be able to understand the free
and principle namely you need to know
things for example about dynamical
systems random
processes you need to know about basian
mechanics some Neuroscience some maybe
computational neuroscience and so on and
so forth so as a as a result it's not
it's an intimidating topic let's say for
anybody who wants to just get started
with it so we seek to demystify FP and
basically provide a comprehensive
framework for Designing neuromatic
models within well where we are actually
trying to mimic humanlike perception
capabilities um so here we present a
road map for implementing these models
and by the way a pytorch code repository
for applying FP in a predictive coding
network is going to be introduced here
the link is here that that's a code it's
publicly available and also as I said
the paper will soon be available um by
the neuro AI Workshop um and so
hopefully you guys will have a chance to
take a look at that as
well now let's get started with the ref
refresher of the math so first foremost
the idea of a PDF so it's an acronym

for a probability density function which is a function that describes the likelihood of a continuous random variable taking on a particular value it has two conditions it has to be positive so for a given parameter θ_0 for example the mean of a gaussian let's say for any input X it's always positive and second we have this normalizer uh sort of condition that if you integrate a PDF across the support space of the input name the all possible values of X you will inevitably must receive get to the actual value of one what this means is the probability of X the input getting any value in a in any range within its support is always one which means that X will always fall in in a range in that in its own accepted range of values so that's 100% so you need these two if you want to have a PDF next is the idea of an expectation I just want to remind you that the reason I'm going through this math is because later on when I introduced the steps of deriving um you know variational free energy these concepts are absolutely crucial so hopefully um I'm not going too fast but um yeah just bear that in mind that these are very important and for those of you who are you know math Savvy and these are two Basics my apologies now the idea of expectation is actually the weighted average of all the possible values that the variable consume weighted by their probability so you have the in the first line um the expected value of x over the probability

distribution the PDF P_X so as you notice
 we're integrating all possible values of
 X multiplied by or weighted by their
 corresponding
 probability and it's important to note
 that you don't you're not limited to you
 know calculating the um the expected
 value with respect to a variable but you
 can also do this exactly the same thing
 with a function so the expected value of
 function is again the same thing but you
 are summing up a weight you're basically
 calculating a weighted average of a
 function where the weights are the
 probability of the input to that
 function which is
 X now some key Probability
 components here now we have this idea of
 a joint
 probability um so a joint distribution
 between two variable X and Y you can
 divide it up between like like some
 conditional term which is the
 probability of x given y multiplied by the
 probability of Y in the first place or
 you can go the other way around the
 probability of Y given x times the
 probability of X you notice that if X
 and Y are independent this conditional
 term reduces to probability of X or this
 conditional term would reduce the
 probability of Y so either way on both
 sides you're going to have the
 probability of x given y um if X and Y
 are indeed
 independent next if want to move from a
 joint distribution to a marginalized
 distribution where you're actually

marginalizing one of the variables out
this is how you can do it so in this
case if you integrate P of XY across all
possible possible values of X in other
words like you literally break this down
and you know basically write it down in
in this you know um constituent terms if
you do this integration you're inev what
basically what you're doing is you're
getting rid of X and then you end up
with a term with a probability
distribution that is only a function of
um

Y and that's what you call a
marginalized probability distribution or
PDF again this is key where will this
come in handy it's going to come in
handy when when I'm going to be talking
about the basium brain hypothesis and
base Rule and so on and so forth so keep
these two equations in mind

please now some properties of logarithms
so this is basically just the definition
of a log so log of a with base b equals
 c the definition says that it just means
 a equals B to the power C right so for
example the log of 8 base 2 equals 3
because 2^3 is 8 so the log of
multiplication is some of the logs of
individual terms the log of a division
is the subtraction of individual log
terms that log of denominator minus the
log of the
denominator um the log of a fraction is
literally minus the log of the swapped
fraction literally swap b and a together
from like you know you end up with just
a negative of the of the new loog

term um the power law the log of B to the power C so basically C drops down as a coefficient so you end up with this term $C \cdot \log_a B$ with base a log of a with base a if the value and the base are both the same you end up with value one and by extension if you have this term C drops down as a coefficient and this term equals one so you end up with C log of one with any base is always equal to zero why because a whatever it is to the power zero is one and last but not least this one is again very

important $\ln$ of a is basically just a logarithm a logarithm with you know where the Bas is natural number e or it's called the natural log and E is basically the oiler Oiler number so you got log of a with base e it's called just the natural log so when in in Python when you say numpy do log is basically the natural

log now the free energy principle it is a theoretical framework in Neuroscience it's been developed by Carl friston

so basically things that persist over time especially living systems look as if they are minimizing a certain quantity it's called free energy it is as if they're uh sort of maintaining this boundary that is called a mark of blanket to persist in the face of random fluctuations in the uh in the environment um and this really connects well with the beian brain hypothesis you as you will see shortly

so the brain from the perspective of
beian brain

hypothesis it's actually in the game of
inferring the hidden causes behind its
Sensations so namely if you have if you
let's say y to denote the sensations and
 X as the hidden State and then this
inference basically means that given my
observation why I want to estimate the
probability distribution over
 X right it's basically called a
posterior probability which means that
after having seen an
observation um what is causing in the
world beyond my Markov blanket this
particular sensation or
observation inevitably this requires
having a generative model GM of the
world and this actually connects really
well with Helm hold's um concept of
unconscious inference and it is defined
like that perception is kind of
unconscious inference which means that
we are in the game of you know inferring
the hidden cause behind our our
Sensations for example right now you're
looking at the Monitor and the photons
but the light is hitting the your
sensory epithelia on your eyes those
excitations those those hits those are
your sensation and when you realize that
I'm I'm watching this presentation at at
at active inference Institute um and
this guy is talking his name is mean
this is the slide of a of a face Helm
holes pictures here these are the
inferences that your brain is making
given that particular Sensation that is

hitting the the neurons on the surface
of your eyes right so it's estimating
this probability distribution about
what's happening beyond your Markov
blanket now let's talk about the idea of
generative model versus generative
process this is actually very very
important on the right hand side you see
this IDE this generative model as a box
and it has some internal States let's
call them hidden

States this represents the neuronal
activities in your brain um the
excitabilities

um different you know the the voltage
the voltage level let's say of your
neurons that are keep changing all the
time they fluctuate all the
time and these guys I BAS basically
holding information about what's
happening beyond your Markov blanket
right remember what's happening beyond
your Markov blanket is out of your reach
you have absolutely no way of observing
exactly what's happening

right and here we have the generative
process where x^* denotes the actual
things that are happening in the world
which are completely out of reach to you
now if you imagine here you have the
face of a person let's say steep job is
standing in front of me but this is no
like I'm I'm not going to see the truth
what I'm going to see is some sort of a
um a map some sort of a noisy
transformation of the truth x^* right
let's call it why this is what I'm going
to be experiencing this is what's going

to hit and uh induce excitation at my sensory epithelia on my eyes and then if you want to look at like um basically dig deep into the idea of perception as um unconscious inference the G the game is this your brain the generative model in your brain predicts a sensation like I'm expecting this sensation to be felt or received at my sensory epithelia on my Markov blanket and then the actual sensation why is received so the difference between the actual sensation and my prediction is called the prediction error so using that prediction error I will update my G model about the world namely I will update my belief about what's happening in the world so this is where my internal States will change to reflect a better image of the unknown World beyond my markof blanket or the parameters of my generative model are going to change um to again do a better estimate of what's happening in the world so that I will have a better prediction about the sensations that that world is going to cause right so generative process again is not directly observable and the internal uh States X this is very important carry information about the invisible external States x beyond my mark of blanket this is where this connects to the IDE the discipline of information Theory this is a very important link by the way so the generative model what do we mean by that mathematically speaking now in a dynamical world a generative model

has belief about the Dynamics of the world what it means is how fast is the world changing for example or how is it changing now if the world is at a given State X right what does its Dynamics look like for example it's velocity or acceleration or even higher temporal derivatives are like jerk basically all sorts of rates of changes the rates of change about the world that's what what I mean by Dynamics so in a generative model your generative model will have an understanding has to have a belief about how the world is changing Beyond a smart C blanket so in other words if I'm in the game of estimating the Dynamics of the world it means I'm chasing a moving Target what it means is at any given time when I want to infer what's happening in the world by the time I've made that inference the world has already moved so that basically makes inference uh a tricky task especially if I'm in a in a you know highly dynamical World it makes it more challenging the second thing that in a Gena of model we should have is the Gena of model has a belief about how the world causes its Sensations namely if the world is at any given state that by my estimation beyond my Markov blanket what is the most likely sensation why that I will be experiencing right so these two are the key components of a generative model so like mathematically this is how you show it $\dot{x}$ is literally the velocity of uh the world how fast it's changing

mathematically it's just $\frac{DX}{DT}$ and then you have some function of the states X and causes new and Theta the parameters of my generative model plus some random fluctuation ΩX so that equation is called a state Dynamics equation but what it basically is telling you is that the Dynamics of the world is driven by X by the position of the world basically and these are all my belief about how the world is changing $\frac{DX}{DT}$ the velocity of the world and technically speaking f is called the flow of the state Dynamics it's the deterministic part of the state Dynamics ΩX random fluctuation is what encodes your uncertainty about your belief about how the world is changing yeah the second equation uh basically connects to the second concept that I introduced here it's basically telling you that my belief about my sensation about like what's the most likely um Sensation that I'm going to be receiving is encoded in this equation where my observation is equal to the G of the function of again hidden States the causes new and the parameters some parameters plus some random fluctuation Ωy it's called the observation model again Ωy encodes your uncertainty about your belief about the most likely Sensation that you will be experiencing a sensation that is caused by the world you notice that we have this these again your hidden saer X and then uh the cause

is new now if you're wondering so
you know what the thetas are these are
just the parameters of the functional
form of FNG G but if you're wondering
what causes are this is something that
I'm not going to be focusing in this um
particular talk but just imagine that if
you have a hierarchical generative model
these causes are just a subset of the
Hidden States these are like the hidden
states that do not have Dynamics but
they're the tools that different levels
in the hierarchy use to be able to talk
to each other so interlayer
communication is done using these news
or the
causes these are the ones that basically
Drive the Dynamics you notice that
that's why we have $x_t = f(x_{t-1})$ and
new they they drive they drive the
Dynamics without them the layers cannot
talk to each other
basically now if you want to like turn
this into a belief-based
system where you have probabilities in
play you just need to assume a
Gaussian distribution over your random
fluctuation namely with mean zero
and some Precision ϕ over your random
variable X and your observation y just
my you know Open brackets Precision is
just inverse
uncertainty um and close brackets that's
just a notion in
statistics Now using this if you
literally replace um you know this um
notion of a Gaussian distribution into
your original equation you end up with

this probabilistic sort of um uh you know notion that the probability of X do given all of these three elements is distributed according to our normal distribution where again the Precision is the same as the Precision of your random fluctuation but you notice that because we're summing up f with this guy over here the the mean of that Gaussian distribution becomes this F term same story for the your belief about your why your your observations and that is why I'm saying that f of x and new and Θ is the expected your expected um Dynamics about X do because it's literally the expected value of the gaussian distribution same story here G of X new and Θ is your expectation about your observation your sensation again it is the sort of expected value of this gaussian distribution the value that is most likely to occur so that's generative model two things we know the Dynamics of the world we have beliefs let's say we don't know it we have beliefs about the Dynamics of the world and we have belief about uh basically the most likely Sensation that we will be experiencing if such and such was the State of Affairs of the world let's say x was the State of Affairs in the world now we talked about inference let's talk about Bayes rule really quickly now here this is base Rule and the components of Base rule are as follows we have the likelihood term which is the probability of my observation given uh

the state of the world we have a prior belief which means that before observing why this is my belief about what the world looks like we have the model evidence which is probability of data and these two together is called basically the the generative model again like this is in line with what what we discussed here this is basically your prior belief about the Dynamics and this is literally the mapping between X and Y which is your likelihood term so I'm not saying anything new here these two together are your generative model and this is your model evidence this is a devil right this is intractable you notice that in a real life scenario we will not be able to calculate this and this term on the left hand side is called posterior probability it's your belief about the distribution over hidden state of the world X and when you map your generative model using base rule to uh this posterior distribution we call this model inversion right so if whenever you hear model inversion in the context of FP that's what we mean by that so this evidence again going back to our idea of marginalizing a variable out if you again just remind yourself that this is how you marginalize the hidden State hit or X out you just end up with probability of let's say sensation why you notice that this integration could get really nasty really fast which means that if you have you you could have basically any number

of hidden States X and for each X you would have a range of possible values so this thing could turn into a potentially like potentially in infinite number of nested integrals basically it can it can become intractable really really quickly so what this means is we will not be able to calculate this posterior probability u_m exactly so what should we do the answer is variational inference so because U you you see probability of data your model evidence is intractable but at the same time ironically it's the true measure of how good your generative model is u_m which means that we call this also marginal likelihood right so if it is high it means whatever sensation you're experiencing right now under your model of the world under your generative model of the world the probability of that particular sensation is very high it means under your model of the world everything that you're sensing you already expected to to sense it so it means you have such a strong and uh you know and precise belief and precise yeah belief about the Dynamics of the world and how the sensations are generated from those uh those dynamics that whatever you're receiving is not surprising that that's why model evidence is such a strong metric for evaluating how good a generative model is but unfortunately it's intractable so can we find a bound to this namely a lower bound and the answer is yes I'm going to introduce the

idea of an evidence lower bound and the next slide so by doing this I'm turning the problem on its head so I'm turning a converting let's say an intractable exact calculation of the

posterior and I'm replacing that with a tractable

approximation um well basically a tractable approximate estimation of the posterior and that's called variational inference and I'll tell you what why we call this variational by the way because the objective function that we're going to be minimizing to do that is VAR free energy is actually a function of a function it's a functional so where wherever you're dealing with functionals and optimizing functionals U that's where the calculus of variation comes into play that's what it means by variational

inference now on the right hand side I'm just going to put some reminders the expected value and the this particular particular law of logarithms and in the middle this is called Jensen inequality just keep keep them keep an eye on this one it just says that the expected value of a log of something is always a lower bound to the log of the expected value of that thing right so this is going to come in handy so

let's study the this U amazing concept of evidence lower bound remember P of Y is intractable we want to find the bound to

it so this is log of probability of data um this is how we mark marginalized X

out and you can multiply and divide by
this by by by something right whatever
it is so here Q_X is we call this
surrogate the surrogate posterior over
the hidden State this is going to be a
representative or your approximation to
the true unknown posterior belief right
so let's call it Q_X and it's by the way
it's really a key factor in variational
inference and so keep that in mind
this Q um um variable this Q
function so moving on you you do a
little bit of play here so this this
sort of Q moves to the denominator and
you notice that this is the definition
of expectation with respect to this
particular distribution Q right just
getting the definition from uh from
here and according to Jensen's
inequality um this term has a lower B
and this is the lower bound to it so the
expected of of this term the expected
value of this term over this probability
distribution you have the joint here and
you have the surrogate distribution here
and you see so this right hand side
again remind yourself this is your log
model evidence and we found the lower
bound to it using Jensen's inequality
right so the lower bound is found this
guy is your evidence lower
bound um and that that's a key finding
that by maximizing this evidence lower
band you're actually maximizing uh your
your log model evidence and you're
getting closer and closer to the true
posterior now elbow and vfe just want to
make sure that we're clear on this we

have elbow here um but VF or variational inference is just negative of your elbow so if you multiply NE minus one by both sides of this you end up with this term always bigger than minus log probability of data this is called surprisal and this is called variational free energy that's why we say VF is always an upper bound to surprise so maximizing elbow or minimizing VF they both mean the same

thing and um this is basically called variational free energy and that's why we call this a functional because this thing is a function of a of another function and this called surprisal or surprise and just shows you in a plot that here you've got variational free energy that's always an upper bound to the true unknown surprisal or by extension you got your elbow here that's always a lower bound to the true um log model evidence so VF has multiple incarnations it comes in different form by playing with the math so if this is your this is your vfe just expand you can just expand the joint and the denominator and then literally use the log rule of the you know uh the division turned into the subtraction and then you can distribute the expectation across the two different terms and this um this basically results into like just basically swapping these two terms uh by each other you end up with this equation right and these terms have names so this one actually this expected value between your belief about the the hidden State

and the true unknown posterior this expectation is called a
 a KL Divergence it measures how different your belief about X is from the true unknown posterior that's why it's called an approximate error approximation error and the second term is your surprise
 right um so um if I want to like show you like what this actually means visually this is really interesting so this is your surprisal right and on top of surprisal you're adding the approximation error this is this sort of length over here and then this constitutes your free energy what this means is like free energy is always an upper bound on surprisal and the error the that Gap is your approximation error is how far your belief about X is from the true posterior so by tuning your generative model your variational free energy goes down and when it goes down this Gap is reducing so which means that your Q gets closer and closer to the true true posterior which means a better and better perception about the world and this doesn't end here your free energy will get closer and closer to the surprisal um basically by extension to the model evidence which is the metric that we're going to be use using for you know comparing different models with one another a second incarnation of vfe is again we're going to expand the denominator again into this joint so previously was $P(x \text{ given } y)$ times probability of y now we go $P(y \text{ given } x)$

times P of X and again Distributing u
 the log and then we distribute the u
 you know the
 expectation and then just you know u
 swapping the terms again you have this
 scale Divergence over here and you have
 this negative negative expectation now
 this thing over here is called accuracy
 it means how accurate am I under my
 belief of the world the expect the
 expected value of observing why under my
 estimation of where the world is X so
 the higher this is it means the more
 accurate your belief about the world is
 that is that is basically causing your
 observations but at the same time this
 term over here is called complexity it
 means that if you have some prior belief
 about the world you you will not be able
 to just move your queue wherever you
 want it is it has to stay close to your
 prior belief over here so this is
 basically a reg regularizer if you
 will u and the third
 Incarnation is u basically again here
 this is the definition of v again you
 know the log rule u so we end up with
 these two terms so this guy over here is
 called expected
 sorry it's called expected internal
 energy and this guy is your entropy so
 this the expected value of log of QX we
 call this entropy
 u how do we minimize raal free energy
 we have to maximize our entropy and we
 have to maximize our internal energy now
 this term makes sense that we need to
 maximize it but what does this one mean

it means that your Q will stay as smooth as possible as simple as possible this connects really well with aom's Razer if you're from machine learning background that your the best model is the one that provides the most accurate account of the data but at the same time the simplest explanation for that data this Simplicity is enforced by this entropy term it it states

smooth so this is very interesting so these were just three incarnations of variational free energy and the goal of inverting the generative model is basically we the GM strives to minimize this vfe um this achieves two things I hinted this uh just a little bit previously but it means that the actual uh value of VF will approach the intractable minus log probability of data this is going to be used for model comparison and the surrogate distribution will approach the unknown posterior belief remember that was the goal of um you know perception in the first place and once you reach that basically minimize your variational free energy you can use the actual value of free energy as the metric as a proxy for the unknown model evidence to compare different generative models together right so the model with the lowest variational free energy is going to be the best model yeah I think I've explained that one now just to let you know that the true form of the posterior is unknown and you see the the the terms like the the probability distribution

The Joint distribution here um it is not guaranteed that um you know they're analytically simple you know to to derive and and you know work with because of that um there is this idea that we can actually use llas approximation to break down each one of the distributions and basically approximate them using a tailor expansion um particularly up to the third term which means that I'm providing a quadratic approximation using tailor expansion for each one of the components um within variational free energy um unfortunately I don't have the time to go over the detailed derivation but suffice it to say that the result of you know s approximating all the components is that the variational free energy simplifies the dise equation for those of you who are interested at to know the details I encourage you to look at the primer on variational plusus paper which is an amazing paper um it goes through all the details of derivation so basically free energy um reduces to log of the joint distribution and you have the posterior um covariance and then yeah n is basically the number of parameters and the states that you have and so on and so forth now predictive coding this this is basically where it gets even more interesting it is a basically a hierarchical generative model that we're going to be using to model human perception and the goal is to invert

this hierarchical model so it's a powerful framework to describe how the cortex accumulates evidence from noisy stimuli um it's an application of the free energy principle it focuses on how this literally fantastic organ can perform probabilistic inference about the most likely hidden cause that has caused the sensory signal um it assumes that the brain has a generative model of the world through which it constantly infers the hidden states of the world and it can also learn the parameters while inferring the hidden state of the world so model inversion happens by minimizing variational free energy again even though it's hierarchical but the concept is stays the same so in predictive coding we have this concept of top- down predictions where predictions go from inside out about your Sensations right and then the prediction error is calculated like what I described before and then this these errors if I want to be more precise the Precision weighted prediction errors are sent back from outside in uh or they call it some up sort of Direction so that the the estimations of the world the estimation of the posterior beliefs about the world are corrected basically that's what model inversion that's when model inversion happens when the posterior estimates get adjusted and it looks like something like this we have this hierarchical model that we have the same story of State Dynamics and observation

model but here we have this happening at each level and then you notice that these news these causes are the ones that link layer l to l minus one that's the important part that without these news without these causes the layers cannot talk to each other you notice that it is a hierarchical dynamical system uh basically um now how do we design an experiment um for you know for for for this kind of perception neuromorphic modeling using the free energy principle I just want to give you the sort of ingredients here you the designer you first have to Define your generative process that's X that's the true uh hidden state I should have maybe said x yeah x star would have been a better notion then you will generate the observations this is some sort of some transformation of the X 's you can add noise to it you can transform it the way you want but that's going to be the observations that's going to hit the Markov blanket this these are what this uh this is basically where the sensory state observe and receive those Sensations you have to Define your generative model this this part is the tricky part right the f 's and G 's in your dynamical system that's where uh everything gets really tricky how would you define them if it's hierarchical how many layers should it have basically these are the big questions you have to define the

evaluation metric that how would you define different Genera of models um with each other and then you uh you have to design the online State inference Loop now this is by the way just for this case for this presentation is just about State inference and then you have to compare different gen of models this is where beian model comparison comes into play now here um you basically use a dynamical system here I'm just using a lotka voltera process that you know captures the Dynamics between uh Hunter and and the prey together it's a two-dimensional sort of world if you will so by solving the trajectories by solving the differential equations you get to this sort of trajectory so you have 1,000 data points here each which is two-dimensional so this is the unknown states of the world right this is the unknown dynamics of the world to gener a process again it's unknown it's beyond the mark of blanket again so this is the figure that we saw before now you can assume that you got the brain here and you've got uh the generative process xstar on the left hand side the observations why are basically a noisy version of the same hidden states that you notice that this this is really noisy here right so the brain has to receive that and then infer the true States here right so for a one layer predictive coding Network for these particular experiments you have just s a simple

sort of generative model it's just one layer and then uh I'm sort of comparing two generative models here where the flow of the of the state Dynamics are different on the left hand side you have something that we call just it's just just a pullback attractor uh we have this 2 by two Matrix here and we have this sort of point of Attraction which assumes that everything collapses into this world just just to be clear it means that my generative model of the world assumes that the world Beyond The Mark of blanket is attracted to a particular Center called saigh right that's what I mean by believing something about the world whereas the second flow the second generative model is a sinusoidal sort of belief it means that I believe there is something periodic about the world beyond my Markov blanket and the G's I just kept them as identity which means that whatever new is is going to be my prediction about my sensation right so I kept the G's identical now the evaluation metric this is very important as the inference is happening online for every observation you have to make an inference um and then predict the sensation and then calculate the prediction error and then eventually calculate variational free energy because it's happening online you can actually literally accumulate your free energy variational free energy across the entire inference period so at the end you will end up with one scalar

value it's called the time integral of
of your variational free energy they
call it free action right it's just a
fancy way of summing up variational free
energy across the entire inference
period in basically it's time
integral and it is
it's an upper bound to the accumulated
surprise so variational free energy is
an upper bound to surprise and um free
action is an upper bound to the
accumulated surprise because free action
is in and out of itself an accumulation
of variational free energy and this is
you have to also then Define the online
State inference Loop uh I don't want to
go into too much detail here but um
basically for each observation you will
make predictions and you calculate the
prediction errors you calculate your
free energy you calculate the gradient
of free energy with respect to your
estimations of the posterior that's the
key part and then this is where you have
to update your beliefs about the world
and you notice you notice that it's a
differential equation because the way
you're going to update your um your
belief about the world it's
continuous it happens in continuous time
it means between each two observations
 Y_i and Y_{i+1} I have to integrate over
this update rule to move from my
previous belief about the world μ_{i-1} to a
new μ_i at the next time step and that's
why the integration happens here and
just so you know this equation is
basically saying that the amount by

which I'm going to change my belief about the world μ is first of all the gradient of free energy with respect to μ right that's typical gradient descent but the second term it serves as a momentum it means that if I'm going to update my belief about where the world is I'm going to consider the velocity of the world as well right if I'm going to estimate the velocity of the world I'm going to consider the acceleration of the world as well it's it's similar to the idea of momentum in optimization algorithms this is the bit that helps you you know catch up with the chasing Target that is the world that I talked about in the beginning and then by you integrate this differential equation to to update your belief about the world and then the next observation happens so this continues for all the observations and you're accumulating your free energy as time goes by and the result will be your μ your free action so the result is this so this is the first generative model with the pullback attractor μ so this is the uh inferred state right and this is the this is interesting this is the accumulated free action right um very very interesting you notice that free action μ is having these regular jumps it means periodically I'm getting surprised about my Sensations remember that if you note here the this is where these nonlinearities happen it means that my simple pullback attractor fails to capture those Dynamics so it gets

surprise but interestingly where things are linear and simple I'm not getting much surprise so it's almost constant here so I'm good at capturing simple Dynamics but not highly not the ones that are nonlinear whereas the sinusoidal one you notice that we are capturing the magnitudes much better we have a steady increase in free action we don't get surprise for some Dynamics and less surprise for other Dynamics we are always equally surprised for everything which is interesting that like here we are getting surprised consistently for everything but here we are not getting much surprise for these particular Dynamics so one is good Capt good in capturing nonlinear Dynamics periodic Dynamics the other one is better in capturing simpler linear Dynamics which is interesting and notice that the reaction here is much lower than the freeaction here so basically this means that this model is much better now looking at the Gen the generative power of these uh GMS so this is the G function if you remember G of X and μ that's the prediction of your one layer predictive coding Network about the sensation that's the bit that gets compared with the true sensation and the prediction error is calculated remember that observations just like h_{den} say are two-dimensional so this is the first dimension of why the true sensation and why had the predicted sensation you notice that they're almost close and also with the second u_m Dimension they

it's not doing a bad job right but for the second generative model again it's doing a much better job in capturing the magnitudes of this of the system again you notice that it's not so good in generating these bits whereas this the first model is better in generating prediction more accurate predictions of the of the first

St now these are the results the pullback attractor and trigonometric the sinusoidal um sort of flow you notice that the freeaction of the second model is lower it's better and M loss which basically comp uh compares the estimated X and the true X the inferred X and true X together is also slightly lower so free action shows that the second model is good and this is where beija model comparison uh comes into play where we use base factor for comparison it basically says that we take the the model evidence for each one of the models and we divide them now because these are intractable we're using free action as a proxy and when you put the values of free action you get to this value because it's higher than one it basically means that you have to pick the second model because it has a lower free action it doesn't get as surprised as the first one and at some point of caution free action versus MSE MSE as a measure of accuracy should never and I repeat never be consulted on its own because it ignores complexity so for example what if the freeaction of model one and model two the model one had a

lower free action but what if the MSE of the first model was actually higher than M than the second model which one would you choose again you have to choose model one because this has higher power of generalizability to the Unseen data so it has less chances of overfitting to the data we call that free action again as complexity minus accuracy so what if we remove complexity again we will overfit but that was important some final thoughts you can definitely learn the parameters and also estimate uncertainty about the world so these are the things that are not covered in this presentation but you can do that this is where the separation of time scales um will come into play the demm algorithm does exactly that this is dynamic expectation maximization I encourage you guys to take a look at that algorithm it's you using that you can solve the triple estimation problem now can I go beyond x dot yes you can that's where you jump into the world of the generalized coordinates of motion where you're not only using X estimating X dot but you're also estimating higher temporal derivatives of uh of the world you have a a system of couple differential equations rather than having just two equations which is very very interesting actually um also remember that active inference is not the same as predictive coding um particularly in the philosophy um you know um communities I've heard that people use active inference when

they're talking about predictive coding
remember that in predictive coding we
are simply doing um making sense of the
world there is no concept of planning or
action in play um the ml Community
unfortunately um does not work with the
Dynamics of the world and AI uh um
basically is going to the opposite
direction instead of uh using smart data
we're just going bigger and bigger data
so we would like a world that chat GPT
would also ask you a question and
becomes curious about you to to receive
information that it selects basically so
where should you start with with the
free energy principle these are the
resources that I strongly recommend you
some videos on YouTube hopefully you can
later on pause this video and go through
these links some papers that I'm
recommending here and the last one the
two books that I believe the red one the
predictive minus for philosophy
community and the first one goes through
the math extensively it's a beautiful uh
book yeah the uh and the videos I I I
recommend that you guys start with the
videos in the first place and I would
like to just acknowledge our funding
from the Maryum and also my colleagues
Simon bbus and professor priston and
also acknowledge the folks that UCL UC
UCD and monach University who have
always supported uh uh our work and
here's the link to the the code that
everybody can just take a look and run
the read me file is self-explanatory
unfortunately I ran out of time and I

won't be able to show a demo but yeah that's it so thank you Maron that was great and and uh we have 10 minutes if you want to show anything or I can ask some questions but but we have till five minutes after if you want to show anything or even just scanning the repo could be cool okay sure sure sure that's great thank you for that but just wanted to comment your your educational material and the YouTube channel are great and and they are they're an excellent supplement to the textbook and to the math learning I appreciate that yes um absolutely uh so yeah I also have a YouTube channel mlon feel free to take a look I'm actually reading the active inference book on that channel so hopefully it would be of interest to you guys and this is the repository that is publicly available um the read me file is pretty self-explanatory and this is just a one predictive coding Network for the task of inference and modeling human perception as a task of inference these are the same pluss that you've seen um so basically all the functions are here right everything you need is here so the main function is the one that you're going to be dealing with and the yaml file is where you actually choose the parameters uh of your experiment just to show you in action what it looks like so if you actually clone this repository on your machine you just have to go to the yaml file and just do a quick scan of the readme file but basically uh using

this is where you define your generative process for example you can say I want it to be a lotka vola process and this is how I'm going to be picking the DT and the total time uh to actually solve the trajectories of my generative process you can Define your generative model here you can Define the Dynamics to be like a pullback attractor or a trigonometric one the likelihood for example is identity here these are the dimensions of your X and Y two and one you might wonder why it's two for X is because you have x and $\dot{x}$ together where the world is and how fast the world is changing and this is just Why in your generative model you can increase them uh that's where you jump into the world of generalized coordinates of motion but this is beyond the scope of my talk today again this these bits are beyond the the scope of our talk today this is where uh you basically C talk about the correlation between the noise process across different levels of the generalized coordinates so the noise on the velocity how does it correlate with the noise process on the acceleration your belief about the acceleration of the world so how do they interact with one another that's where these um values come into play and then this is the kind of noise that you will use to alter the true states of the world and generate your observations for example you can use a colored noise here you define the mean and uh basically the standard deviation

of noise and this is we're using the sort of Kernel here that convolves with the white noise to actually generate a more smooth noise the reason being that um the biological systems are believed to be uh working with uh you know to so all all the signals that you work with in biological and organic systems are believed to have been generated by some sort of dynamics that that's why we keep them as smooth noises infinitely differentiable noises rather than just white

noise and uh if you actually run this so this is basically a pullback attractor this is the first uh sort of generative model that we worked in so you just run this main

function so we have 1,000 observations noisy observations right

and immediately this happens so for every 10 observations we are accumulating free action you notice that free action is ever increasing because it just accumulates free energy and on the right hand side you have VF or variational free energy um again if you look at the definitions inside uh the functions folder of how we calculate or compute variation of free energy you notice that um it's a far more simplified version of what you would see about energy is because everything is under llas approximation and you end up with far more simpler terms and the final result is reaction is this much and basically your final MSE loss is 0.53 and if you want to see the final output

you just have to go into the results folder and these are the folders that are generated just so that you have two dimensions in x one dimension in y lot voltera is a generative process for example do generative model follows a pullback attractor scheme and then your G is just identity this is your free action and then that is your M_s Loss and if you open it all the plots that I showed you in the presentation are saved as PDF here just to give you an idea like this is your X this is your true invisible uh states of the world right and if you want to look at exat this is the inferred states of the world and these are for example the the noisy observations of the world yeah so I hope that the repository would be um of help and of interest to all of you thank you very

much great what kinds of problems or settings do you feel that this could be adapted to or useful

for um I would say first foremost this could be used for um basically it's it's all about any kind of problem that's about inference you have some noisy observations that you want to infer the hidden state but in terms of adaptability you can build on that by introducing parameter learning as well as uncertainty estimation so our goal is to develop this into a triple estimation sort of package predictive coding Network and add more layers to it so to have a hierarchical predictive coding uh capable of solving the triple estimation

problem yeah that's the ultimate thing
that you can actually adapt this
cool
um what are your uh what are you excited
about for ML Dawn or for your learning
or research in next
year um I'm very much excited in
embedding these kinds of calculations
into the world of AI because as I said
AI is moving in my opinion and my humble
opinion is moving to the towards a wrong
direction instead of bombarding models
with all sorts of data that the model
should be able to choose its data so I'm
very excited in you know developing this
framework into a hierarchical predictive
coding Network and then sort of compare
the performance of such a model with
other traditional aib based generative
models and basically show its
superiority um that I believe it has in
different scenarios and ultimately this
model is going to be used for in my
grant proposal and a project I'm working
on is going to be used for modeling
hallucinations as a kind of false
inference about the world um and it's
going to be used as a proxy to for us to
understand what happens in uh
Neuropsychiatric disorders when people
actually suffer from hallucinations in
the brain so this predictive coding
network is going to be a brain inspired
AI model for us to understand what goes
wrong in in the skull of a human being
when they suffer from hallucinations
about the world so that's what I'm
really really excited about and really

hoping to achieve cool it's like
inspired by the brain for the
brain absolutely thank you on great great
presentation and good luck with the
continuation of the work appreciate it
thank you for having me okay till next
time e